CSE 421: Lab 2 - Task 2

Introduction to Packet Analysis

Introduction:

In real life, network monitoring and packet analysis is of great importance to observe the network state and identify any anomalous network event. Wireshark is a tool that can be used to monitor our network and also analysis traces of past network events. In this lab, we will get familiar with the wireshark interface and learn basic packet analysis.
In networking research and protocol testing, simulating a network is one of the core works. We can try out existing networking protocols and measure their performance. Also, we can devise our own modified protocol and test its performance against the existing ones. In this lab, we will also get familiar with an open source network simulator, ns3. We will learn how to install the simulator and run a basic simulation.

Objectives:

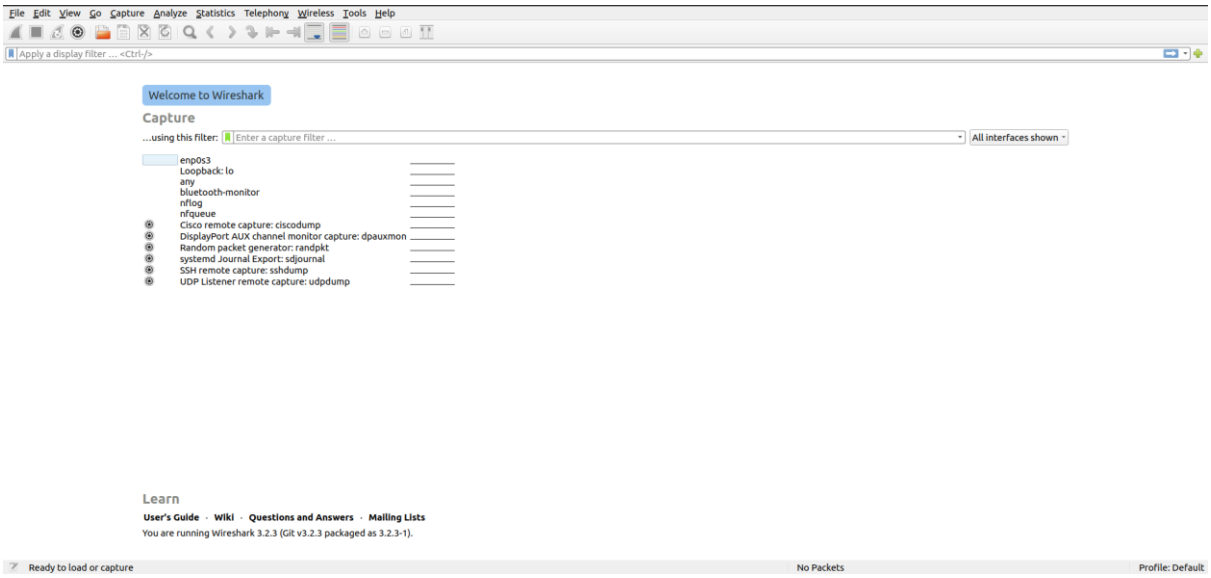
1. Getting familiar with wireshark
    a. Learn about basic wireshark interface
    b. Basic filtering of packets
    c. Explore a simple **HTTP** stream
2. Getting familiar with network simulation
    a. Installing and running ns3
    b. Understanding a simple simulation script
    c. Visualizing our simulation
    d. Collecting network traces and observing the events

Task: Using Wireshark [Do this task in Windows]

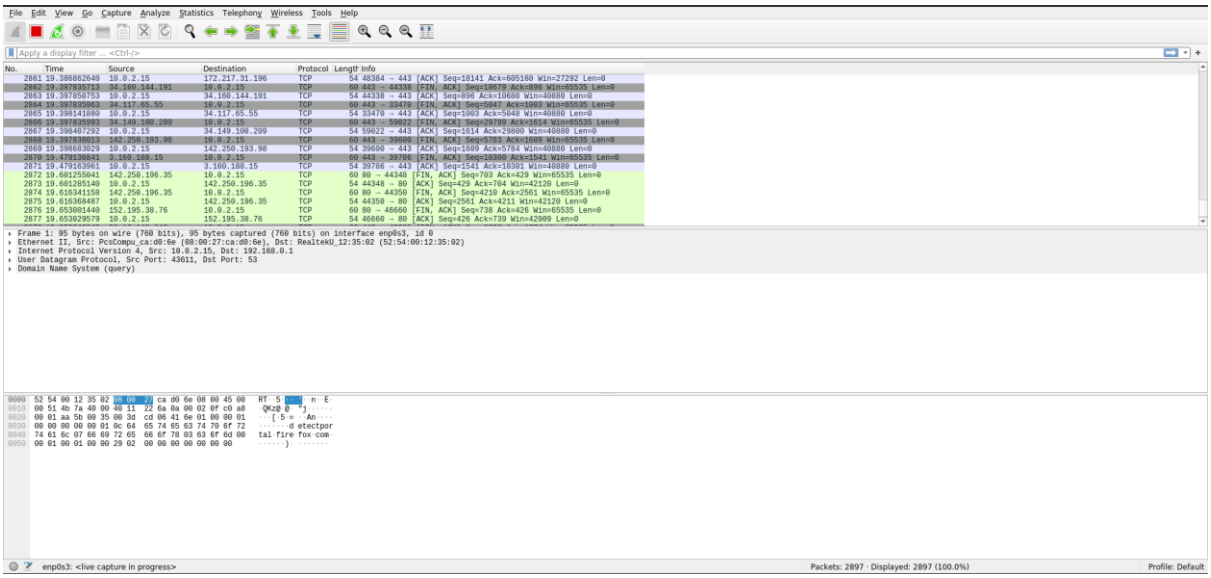
1. Step 1: Download and install Wireshark
    a. Go to https://www.wireshark.org/
    b. Download and install the tool for the operating system you are using

2. Open Wireshark if it is installed

a. You will see a window like the following



b. You will see a list of network interfaces on your device. We will select our Ethernet interface/ wifi interface, whatever we are using. In our case, we will select **enp0s3** and start capturing. To start capturing, we can double-click the



intended interface on which we want to capture the packets. We will see an interface like the following:
Here, there are three sub-windows. The top sub window shows us the list of packets. The middle sub window shows us the headers and data of each layer of any selected packet. The bottom sub window shows us the raw bytes that each packet contains and also the decoded form of it.

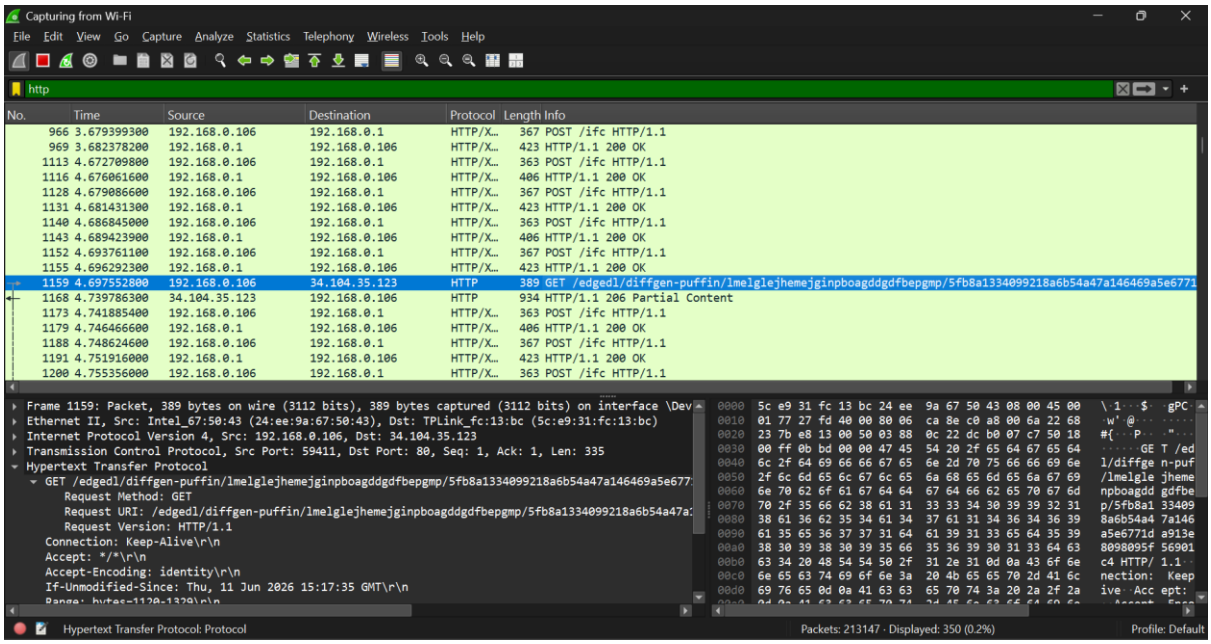
c. We can observe that there are a lot of things going on. That means our device is interacting through the network interface. We can focus on things by filtering out packets that are of our interest. For example, if we want to filter out HTTP data, we can type 'http' in the text box at the top and press enter. Only the HTTP packets will be shown, and other packets will be filtered out.

Home Task:

Visit any HTTP website while monitoring the network traffic. Now filter only the HTTP packets. Observe the headers of any HTTP request and response packet and explain how the layers are operating in this case.

Part 1: HTTP Request

The HTTP request is sent from the client to the server using GET/POST method. It contains information such as requested URL, host, and browser details. This request is processed through TCP/IP layers before reaching the server.



Part 2: HTTP Response

The HTTP response is sent from the server to the client after processing the request. It contains status code like 200 OK and the requested web content. This response is delivered through TCP/IP layers back to the client.

